Deutsche Bank Research Institute

The Brilliant World of FX: A deep dive into restricted markets

4 March 2026

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In this report, we discuss "restricted markets," defined as emerging markets with current or capital account restrictions, or both. We examine their market structure and challenges of trading. Recent regulatory and technological changes offer new opportunities for market participants involved in these markets.

FX trading is increasingly electronified, particularly in the spot market. However, the electronification of FX remains fragmented in restricted markets. Foreign access to local markets is difficult due to regulations and fixed trading hours, often requiring manual FX processes. A common rule requires FX trades to be linked to underlying local investments. Consequently, much of foreign exchange spot execution is handled passively, **often via custodian standing instructions**. Many of these currencies also trade offshore via non-deliverable forwards (NDFs). Despite these complexities, local market regulations are constantly changing, fostering a more competitive FX execution environment. South Korea, for example, is liberalizing its onshore markets to increase FX flows, extending trading hours, and allowing foreign investors access through offshore banks during offshore hours. Thailand's Non-Resident Qualified Company (NRQC) scheme provides economic benefits to non-resident companies through access to onshore Thai Baht (THB) liquidity and cheaper hedging rates.

Adapting to changes in local regulations and leveraging technology, the execution space is evolving to provide foreign asset managers and companies with alternative routes for executing FX in restricted markets. Where possible, foreign asset managers may execute trades directly with the global custodian or a third-party FX bank i.e. not via standing instructions. They are responsible for executing the trades and providing documentation. While **direct dealing** offers greater control, it can be complex and may not provide access to a diverse pool of liquidity.

Outsourcing FX solutions is emerging as a third approach. Specialised FX providers, especially those with an onshore presence, offer expertise in navigating complex regulations and local market practices. **They provide services like efficient order execution, access to diverse liquidity pools, and efficient risk management.** Their offerings can potentially help foreign asset managers and companies achieve better pricing through wider access, and better risk management, for example, via a closer synchronization between the funding and hedging trades. Given that each market is different, there is no one-size-fits-all solution for execution. Technology continues to adapt to this dynamic environment, offering foreign asset managers and companies alternative routes for executing FX..

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Introduction

FX trading has become significantly electronified especially in the spot market, as discussed extensively in our research. Currencies of open economies with open current accounts, such as the G10 pairs, are traded through dozens of venues, and overall electronification has compressed spreads across these markets. However, the electronification of FX markets is more fragmented among emerging market currencies that face restrictions on their current account, capital account, or both. These subsets of emerging market currencies constitute restricted markets.

Many of these currencies have a strong presence outside their local markets, meaning they are heavily traded in offshore markets via non-deliverable forwards (NDFs). Over the past decade, BIS surveys reveal that an average of 70% of forwards in these restricted markets have been traded via NDFs. Furthermore, Asian currencies account for approximately two-thirds of global offshore currency forward (NDF) trading. NDFs on these currencies trade in venues similarly to our developed FX market execution landscape. Overall, the NDF market accounts for 4% of the $10 trillion FX market.

Trading NDFs can also present unique regulatory challenges. For example, NDFs are subject to margin requirements such as the Uncleared Margin Rules (UMR). In our note last year 'FX clearing today' we highlighted that UMR has indirectly increased the cost of non-cleared FX derivatives.

Foreign access to markets locally or **onshore markets,** however, is challenging due to regulatory demands and fixed trading hours, often requiring manual steps in the FX process. A common regulation mandates that FX trades be linked to underlying local investments. Our latest EM handbook discusses in detail the regional regulations and highlights that outside of FX dealers (market-makers) a majority of trading is supported by asset managers and corporates with an underlying local investment.

This paper examines the evolution of FX trading in restricted markets. It highlights the role of non-deliverable forwards (NDFs) in facilitating currency trading outside local jurisdictions and analyzes their impact on forward pricing in both onshore and offshore markets. The discussion then moves to the efforts made by various markets to support the onshoring of FX flows. Finally, the paper explores the electronification and execution of FX, the inherent risks associated with trading in these markets, and their future outlook.

Role of non-deliverable forwards in restricted markets

NDFs (Non-deliverable forwards) offer a way to gain exposure to a currency's performance without the need for funding in the underlying currencies, unlike traditional deliverable forwards. These contracts settle the difference between a pre-determined exchange rate and the spot rate at maturity (either an official domestic rate or a market-determined rate) through a single US dollar payment.

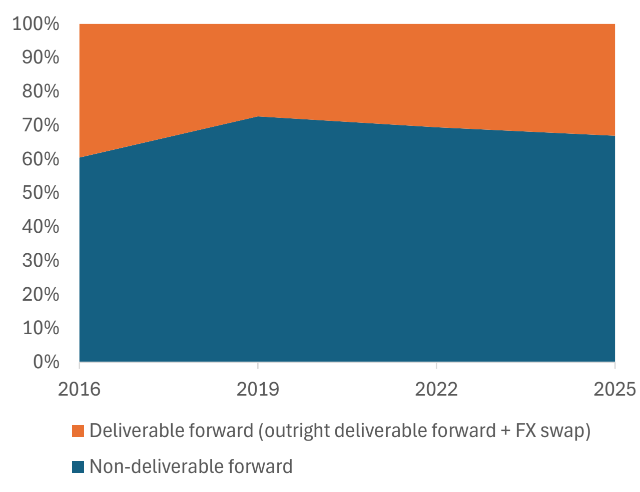
Trading predominantly offshore, NDFs enable investors to circumvent restrictions within a currency's home jurisdiction and bypass limitations on offshore delivery. Beyond hedging currency risk for direct and portfolio investors, NDFs also attract speculators (Ma et al, 2004). The presence of banks and firms with onshore and offshore operations helps to reduce discrepancies in forward rates.

In restricted markets, onshore forward trading typically concentrates on outright deliverable forwards and deliverable FX swaps, while offshore trading specializes in NDFs. The deliverable forward market in these regions also tends to be smaller than the non-deliverable currency forward market, largely due to limited access and shorter trading hours within local currency markets.

Over the past decade, BIS surveys reveal that around 70% of total forwards (deliverable outright forwards + FX Swaps + NDFs) in these restricted markets have been traded via NDFs. Furthermore, Asian currencies account for approximately two-thirds of global offshore currency forward (NDF) trading. Consequently, foreign institutional investors and corporations hedging exposure in such markets have historically relied on NDFs. However, this reliance introduces a potential risk, also known as basis risk: the hedger is unlikely to execute the underlying spot transaction at the same rate as the fixing rate used to settle the NDF. **Hedging with NDFs retains exposure to basis risk between the fixing rate and the rate on the underlying spot transaction. This risk tends to be even higher if the two trades are not synchronized as closely as possible.**

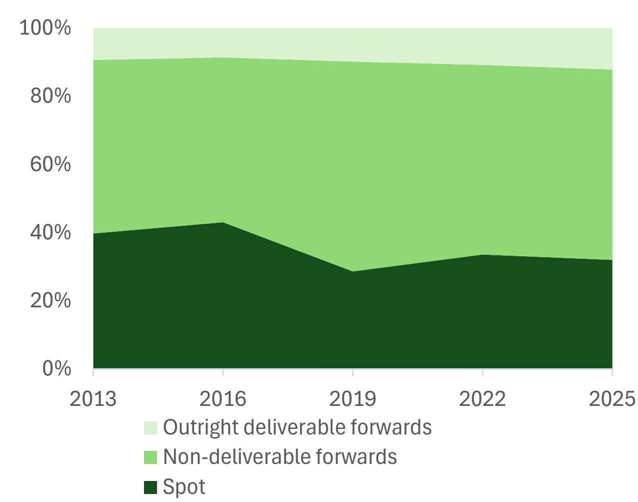
The NDF market's significant size relative to the underlying spot market highlights a strong desire among offshore participants to hedge and speculate on and engage with the currency dynamics of these restricted economies. However, the continuing existence of the NDF market alongside deliverable forwards exacts a cost in terms of lower liquidity from the division of the forward markets.

Figure 1: Forward trading dominated by non-deliverable forwards



Source : Deutsche Bank, BIS. Restricted markets: USD against INR, KRW, TWD, BRL

Figure 2: Asia: NDF volumes have outstripped underlying spot volumes



Source : Deutsche Bank, BIS. Restricted markets: USD against INR, KRW, TWD.
Outright deliverable forwards = Outright forwards - NDF

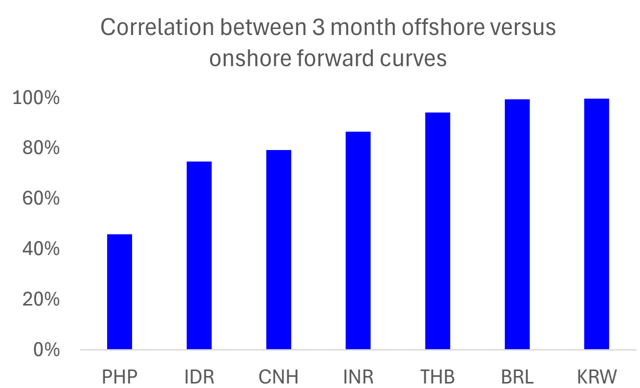
The evolution of NDF markets and impact on pricing

The evolution of NDF markets has not been uniform, and this has had varying impacts on the pricing relationship between onshore and offshore forwards across different countries. In cases where countries have cautiously maintained restrictions, NDF markets have grown significantly. However, the exact impact on the pricing gap between deliverable ('onshore') and non-deliverable ('offshore') forwards has varied by jurisdiction. For instance, in Korea, local banks' ability to trade both the onshore forward and NDF markets has kept this gap narrow.

Differences between deliverable forward and NDF rates generally reflect the effectiveness of capital controls. A BIS study found that deviations between onshore and offshore curves were largest for the Renminbi, Indian Rupee, Indonesian Rupiah, and Philippine Peso. Previous studies on the Korean Won, Indian Rupee, and Indonesian Rupiah also demonstrate a two-way influence between deliverable forwards ('onshore') and NDFs ('offshore') during normal times. However, during more volatile periods, the NDF market tends to drive the domestic market (Misra and Behera (2006), Kim and Song (2010), Behera (2011), Cadarajat and Lubis (2012), and Goyal et al (2013)). More recently, Schmittmann and Teng (2020) noted that during the COVID-19 pandemic, there is some evidence of NDFs leading onshore markets for a few currencies.

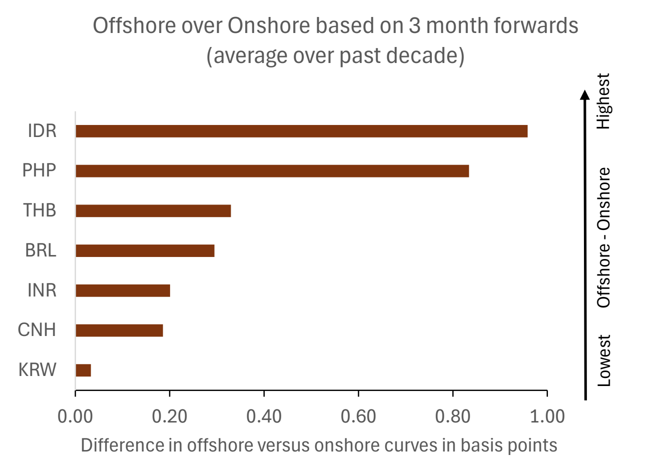
Our analysis also shows lower correlation between onshore and offshore rates for PHP, CNH, IDR, and INR compared to KRW and BRL. [1] Over the past decade, onshore curves have, on average, **traded 40 basis points** below their offshore counterparts in the three-month tenor across currencies(Figure 3 and Figure 4). **This suggests that non-residents investing in restricted markets can potentially achieve lower currency execution costs when accessing onshore markets.** The potential savings can be even higher as these deviations tend to widen during periods of stress, with the BIS study highlighting larger pricing differentials during the financial crisis and the May-August 2013 selloff in emerging market assets.

Figure 3: Deviations between onshore and offshore curves highest for PHP, IDR, CNH and INR



Source :Deutsche Bank. Asian pairs compared at 3 pm tokyo time for spot, onshore and offshore 3M forward. USDBRL taken at 6 pm London time.

Figure 4: Onshore curve, on average, 40 basis points below offshore curve



Source : Deutsche Bank. Offshore(Onshore) = (F-S)/S. Both of F and S taken as USD/CCY. Asian pairs compared at 3 pm tokyo time for spot, onshore and offshore 3M forward. USDBRL taken at 6 pm London time.

Beyond local market restrictions, global factors also influence both deliverable forward ('onshore') and non-deliverable forward ('offshore') rates. For instance, previous studies indicate that both onshore and offshore forward rates increase (i.e., currencies depreciate) when the VIX rises. Notably, global factors appear to have a more pronounced impact on NDF rates for the Renminbi, Indian Rupee, and

1 For Asian currency pairs, we compare the three-month onshore versus offshore curves at 3 PM Tokyo time, and for BRL, at 6 PM London time.

Indonesian Rupiah.

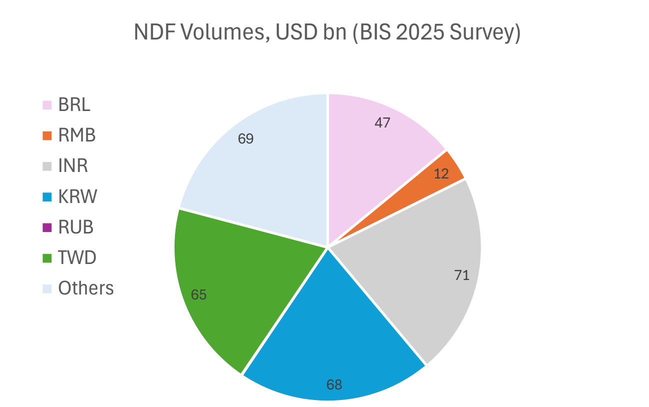
A special note on the Renminbi: its internationalization follows an idiosyncratic route. Chinese authorities engineered a system allowing a substantial pool of Renminbi to accumulate and be freely traded and delivered offshore, all while maintaining capital controls. This strategy has resulted in a tripartite structure for the Renminbi forward market: an onshore market (dating to 2006), an offshore NDF market (established in the 1990s), and an offshore deliverable, or CNH, market (emerging in 2010).

Lastly, when non-residents gain domestic access to forward trading, effectively allowing them to lend and borrow the local currency, the NDF market could become largely unnecessary. Its disappearance may not be immediate, however, as illustrated by the Australian dollar NDF, which persisted for several years after deliverable forwards became available in 1983.

Onshoring of FX Flows: leaders and laggards

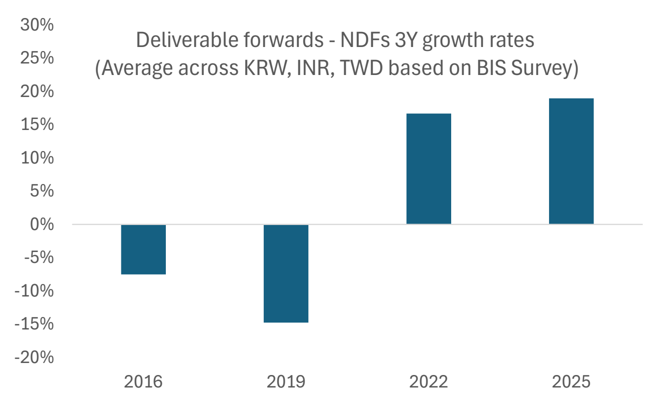
The costs associated with a larger offshore market, including reduced onshore competitiveness, amplified spillover effects during periods of offshore volatility, and a lack of transparency and oversight for domestic regulators, have prompted a re-evaluation of strategies by regulators. As our colleagues Mallika Sachdeva and Bryant Xu have previously noted in their study Asia Thematic Strategy Notes: Onshoring FX flows: regulation is turning the market tables, various Asian markets have adopted distinct approaches to encourage FX onshoring. This should help non-residents gain easier access to the more liquid onshore FX markets and potentially reduce the cost of executing FX trades on the back of their investment in restricted capital markets.

Figure 5: Breakdown of NDF trading across markets based on latest BIS survey



Source : Deutsche Bank, BIS

Figure 6: The past decade has seen Asia outright deliverable forward flows outpacing non-deliverable forwards



Source : Deutsche Bank, BIS, USD vs KRW, INR, TWD

Historically (2013-2019), offshore markets generally outpaced onshore growth. However, the last two BIS surveys (2019-2025) reveal a significant shift: onshore forward trading, particularly in outright forwards in KRW and INR, is now expanding considerably faster than its offshore counterparts. This development potentially suggests the effectiveness of various market initiatives within these regions. In contrast, outside of Asia, some Latin American countries and, more recently, frontier markets are still in the nascent stages of addressing similar market development issues.

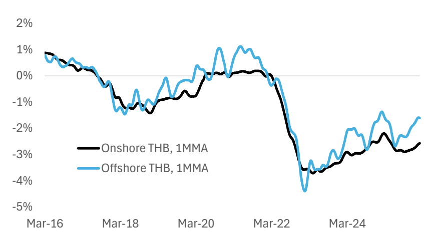
Below, we summarize the distinct approaches taken by some restricted markets to encourage FX onshoring, examining the evolution of both their onshore (within) and offshore (outside) markets.

Korea announced significant reforms to align its FX market with global standards. These reforms include opening the onshore interbank FX market to **foreign financial institutions (RFIs)**, allowing non-residents to trade Onshore KRW FX (Spot, Forwards and Swaps). Onshore FX trading hours have been extended until 2:00 AM local time, enabling asset managers to trade KRW without pre-funding transactions. The scope of RFI transactions will expand to cover all transactions, including FX for current transactions like import/export payments and salary remittances, and Third-Party FX (TPFX) will be promoted. Improvements in FX brokerage infrastructure, including aggregators, will allow corporate end-users to select the most favorable prices. Deutsche Bank was the first bank to execute a trade under the Registered Financial Institutions (RFI) programme. According to Foreign Exchange Trading Trends of Foreign Exchange Banks in 2025 the daily average FX trading volume of foreign exchange banks reached $80.71 billion, a 17.0% increase from 2024, marking the highest level since 2008. As per the document, this growth is attributed to extended trading hours and increased overseas securities investment by residents and domestic securities investment by foreigners.

***Registered Financial Institutions (RFIs)**. RFIs are Foreign FIs that are not based in Korea (Headquarters of foreign FIs, overseas branches of foreign FIs, and overseas branches of local FIs) that are allowed to directly participate in the onshore interbank FX market, if they obtain registration from the Korean authorities.*

India has focused on liberalizing onshore risk management rules, providing market makers with more flexibility for hedging products, and allowing NDF market participation from GIFT City. This strategy has successfully redirected NDF flows onshore, enhancing RBI oversight and aligning forward curves. **Thailand** introduced the Non-Resident Qualified Company (NRQC) scheme to integrate its onshore and offshore THB forward curves, allowing non-resident corporations to access onshore THB liquidity and hedge freely. The NRQC scheme provides economic benefits, as the onshore THB curve has historically traded **35 basis points** below the offshore curve, and the onshore market offers greater depth in longer-dated forwards. **China** is actively internationalizing the RMB by promoting its use in cross-border trade and investments. The PBOC has focused on improving cross-border RMB settlement through international cooperation, enhanced infrastructure like the CIPS payment platform and extended onshore FX trading hours, and corporate empowerment initiatives such as cash pooling programs and increased flexibility in Panda bond financing. **Indonesia** introduced the IDR settled Domestic Non-Deliverable Forward (DNDF) in 2018, traded onshore and auctioned daily, which maintains hedging market access, aids Bank Indonesia intervention, and influences offshore NDFs.

Figure 7: The onshore THB curve generally cheaper and stable



Source : Deutsche Bank, Bloomberg Finance LP

Brazil operates extensive onshore derivatives exchanges for FX, interest rate, stock, and commodity instruments, with Brazilian FX futures and options being non-deliverable and settling in domestic currency. The latest BIS study notes a relatively lower proportion of BRL FX derivatives trading offshore (compared to INR, KRW and COP), with a significant volume occurring via futures on the Sao Paulo exchange, partly due to limited restrictions on non-resident participation. The

Brazilian legal and regulatory framework significantly influences this derivatives market. Constraints on over-the-counter (OTC) trading and a tax system favoring net profit/loss calculations on exchanges encourage trading migration to these platforms.Last year, Brazil announced substantial regulatory changes affecting financial and capital markets taxation – the IOF tax – which increased the levy on financial transactions.

Argentina has removed the minimum holding requirement for non-resident investors in local debt markets and eliminated cross-restrictions between official and parallel markets. The 48-hour advance notice for large FX purchases has also been removed. Additionally, the BCRA has issued a new 3-year FX-denominated bond (BOPREALs) to address legacy commercial debt and dividends accumulated during capital control years. These measures are expected to catalyze further foreign investments.

Colombia has a liquid, competitive, and efficient spot FX market by international standards, though its hedging market is predominantly characterized by NDFs. This prevalence of NDFs is primarily due to historical capital control measures and regulatory constraints inhibiting non-residents' COP settlement, hindering the development of deliverable forwards and swaps. Between 2016 and 2022, offshore trading averaged around USD 7 billion, significantly surpassing combined onshore and cross-border trading of approximately USD 4 billion, while a financial transaction tax further renders deliverable derivatives more expensive than non-deliverable ones.

Frontier markets, as defined by the International Finance Corporation, are smaller markets with foreign investor access limits but offer promising investment opportunities, characterized by early stages of capital market maturity and often lacking financial depth and a shallow domestic investor base. The MSCI Frontier Markets Index includes large and mid-cap companies across 28 such countries, with top constituents like Vietnam, Romania, Morocco, Kazakhstan, and Slovenia. This lack of financial depth results in challenging local market liquidity conditions where direct local spot settlement can be illiquid. Consequently, investors frequently utilize NDFs to gain exposure without needing local accounts. Accessing these markets, especially for African currencies, often necessitates specialised institutions and local correspondent networks due to these characteristics.

For a detailed discussion, please see the Appendix.

Electronification of FX in restricted markets

Our discussion so far has focused on how regulations have shaped the FX trading landscape, historically leading to fragmentation of the forward market. Now, we turn our attention to the electronification of FX markets within this complex environment. While technology has revolutionized foreign exchange trading, boosting efficiency and transparency in G10 markets, the electronification of FX in emerging markets remains fragmented due to restrictions on capital, current accounts, or both.

Restricted markets, such as India and South Korea, have developed localized electronic FX trading platforms. While these platforms remain closed to foreign participants, they are as liquid as some of the G10 FX markets. These platforms facilitate electronic pricing via domestic data and APIs but foreign access is challenging due to regulatory demands and fixed trading hours, often requiring manual steps in the FX process. These challenges also limit the number of FX counterparties providing comprehensive coverage.

A common regulation, for instance, mandates that FX trades be linked to underlying local investments. Consequently, a significant portion of foreign exchange execution in restricted markets is handled through passive products, such as custodian standing instructions.

Despite these complexities, local market regulations are constantly changing, and the FX execution space is evolving, fostering a more competitive environment that could potentially lead to lower costs. For example, as discussed earlier, South Korea has been liberalizing its onshore markers to increase FX flows by extended trading hours and enabling foreign investors to trade Korean assets more freely.

As restricted markets continue to evolve, comprehensive FX outsourced solutions are becoming more common in the most prevalent restricted market currencies. This trend sees more market participants exploring third-party FX execution solutions with restricted currency offerings. We now discuss the different avenues of execution in detail.

1. Custodians

Custodians handle the settlement, safekeeping, and reporting of client assets, which includes managing cash balances and executing FX transactions. They act as facilitators for financial transactions, providing institutional investors with centralized access to their global investments. For cross-border securities transactions, global custodians often provide FX services, including executing foreign exchange and processing tax reclaims.

A key method custodians use is "**custodian standing instructions**" or "**passive products**." These are pre-agreed arrangements from clients that authorize the custodian to automatically execute necessary FX transactions, for instance, when purchasing securities. Custodians have historically been the main FX execution model. In recent years, there has been a strong push from asset managers for more transparency around transaction cost analysis and execution rates and time. Some custodians have begun to provide detailed analysis while others have lagged. In some cases , custodians have also partnered with specialised providers to navigate complex restricted markets.

2. Direct dealing

Direct dealing refers to asset managers executing currency trades directly with the FX execution desk of global custodians or third-party FX banks. Third-party bank refers to an FX bank that isn't their global custodian. In this scenario, the asset manager is directly responsible for executing the FX transactions. This differs from methods like 'custodian standing instructions,' where the custodian executes FX trades on behalf of the asset managers.

Managers may use this method for the enhanced control and clarity it offers over their currency trade as they directly execute the trade. However, direct dealing can become more complicated especially with multiple global custodians. Managing several custodians increases operational burden. Each custodian often has its own FX bank network, with varying requirements and potentially different FX rates. A major challenge is meeting the disparate demands for security details (like ISINs, account numbers) from each FX bank or custodian. It gets even more challenging when a manager trades directly with an FX bank that isn't their global custodian.

Direct dealing means the asset manager must take full responsibility for routing any security and account details, and sometimes for managing tax issues. This also involves bearing the operational and execution risks. Although direct dealing has become more transparent and flexible over the last decade, it still faces significant challenges. Additionally, asset managers may not have access to a diverse liquidity pool that could potentially lead to better rates. This situation has encouraged the development of other execution methods in these markets, as managers look for more reliable ways to control their workflow.

3. Outsourced FX

Outsourced FX solutions involve delegating foreign exchange transaction management to a specialised external provider. These solutions aim to make FX trading more efficient, transparent, and compliant, particularly in complex markets. Technology-enabled solutions are designed to manage many stages within an FX trade lifecycle that are often manual and repetitive. These solutions can also be tailored and provide a degree of control to the user.

While many global custodians offer their own outsourced FX services, specialised third-party providers also exist. Some solutions are "custodian agnostic," supporting accounts held with various custodians.

For restricted markets, outsourced solutions provide a compelling alternative. These external providers often possess expertise in specific market regulatory requirements and have established trading relationships with multiple onshore counterparties in restricted markets. This makes outsourced FX solutions particularly beneficial for asset managers seeking to streamline operations, reduce costs, manage risk, and improve transparency in trade execution. These programs help achieve execution quality and consistency across multiple providers. The outsourced provider manages specific market regulatory requirements and can flexibly navigate across different providers and currencies.

Managers can also benefit from a customized execution model tailored to their operational workflow and needs, effectively outsourcing the complexities of market regulatory requirements and the settlement process**. Such fully outsourced FX programs are gaining significant traction in restricted markets due to their ability to address challenges and deliver benefits like cost savings, improved benchmarking, enhanced risk management, and greater transparency in trade execution.**

Hedging via onshore markets can result in lower costs, with onshore forward curves typically trading below the offshore curves. In other words, the onshore and offshore markets may not always be aligned and the ability to switch between them may be another area of cost efficiency. Conversely, for hedging via offshore markets, higher efficiency gains can be achieved with greater synchronization of prices between the onshore spot trade and the corresponding offshore hedging trade. **Specialised FX providers could potentially help managers optimize their FX process on both fronts.** Overall, the automated workflow should allow managers to focus more on their core strategies, and demand for full outsourcing is expected to grow as FX preferences and approaches evolve.

Given that each market is different, there is no one-size-fits-all solution for FX execution. Some asset managers seeking more control might prefer direct dealing, while others may look for a more automated approach through custodians or outsourced solutions. Ultimately, the decision to choose one over the other will depend on an in-depth cost-benefit analysis.

Non-deliverable instruments in offshore markets

We now briefly discuss the execution space for trading of non-deliverable instruments in offshore markets. These instruments, especially NDFs, tend to account for 70% of the forward trading in restricted markets.

While spot remains the most electronified segment, electronic trading in forwards, particularly NDFs, has been catching up at an increased speed. Like spot, trading in non-deliverable instruments utilizes similar execution venues as discussed in our earlier paper, including single-dealer or multi-dealer platforms, as well as primary and secondary, or alternative venue.

Trading NDFs can present unique regulatory challenges. For example, NDFs are subject to margin requirements such as the Uncleared Margin Rules (UMR). In our note last year 'FX clearing today' we highlighted that Uncleared Margin Rules (UMR) have indirectly increased the cost of non-cleared FX derivatives, incentivizing a shift toward central clearing. It affects buy-side participants, by increasing the margin requirements for uncleared derivatives that mainly constitute NDFs and options; UMR might require a firm trading NDFs to post an initial margin of Y% of the notional value. This added cost is prompting some market participants to explore central clearing solutions even for FX derivatives, despite the lack of a mandate.

Clearing houses like LCH ForexClear clear approximately 25% of the non-deliverable market. Among non-deliverable currencies, the growth in NDF clearing appears to be driven by USD/INR followed by USD/TWD and USD/KRW.

Risks surrounding FX trading in restricted markets

In case of trading in open markets, we outlined several risks surrounding FX trading. Here, we dig deeper into risks that can be potentially different or elevated in these markets.

Non-synchronized custodian cut-offs are a known complexity in the settlement process. Each custodian operates with its own internal procedures, technology, and staffing, which directly influence their capacity to process instructions and meet market deadlines. Clients must therefore be acutely aware of their custodian's specific deadlines to avoid potential issues such as failed trades, pre-funding requirements, or the need for bilateral settlements, all of which **introduce additional risk and cost**. This challenge is particularly acute and impactful in restricted markets due to their inherent characteristics, including higher volatility, stricter regulations, limited operational windows, and more frequent pre-funding requirements.

Another important risk to consider is **regulatory risk**, which arises from changes in regulations or their interpretation. As highlighted earlier, local market regulations are constantly changing. For example, India changed its equity settlement from T+2 to T+1. In March 2024, SEBI announced it would introduce a beta version of a T+0 rolling settlement cycle on an optional basis, which would run in parallel with

T+1. This may in the future mean INR becomes fully pre-funded.

Looking closely at NDF markets, they are advantageous for hedgers and investors because they involve major international banks as counterparties, enabling them to avoid direct credit exposure to emerging market entities and thus **lowering credit risk**. However, NDFs typically exhibit wider and more volatile spreads—especially at longer tenors—compared to deliverable forwards of major currencies, which can indicate higher liquidity risk. For instance, onshore curves have, on average, traded 40 basis points below their offshore counterparts, underscoring this difference.

Hedging with NDFs also entails exposure to **basis risk**, which is the difference between the NDF's fixing rate and the rate of the underlying spot transaction. This basis risk can be further exacerbated in restricted markets by operational challenges. A prime example is the critical need for precise synchronization between a funding trade, such as a spot transaction, and its corresponding NDF hedge. This synchronization is particularly crucial given the fixed trading hours and complex regulatory frameworks often found in these markets. Even a minor delay can lead to significant losses, especially during periods of market volatility or illiquidity. Therefore, accurate timing is essential to prevent unexpected price changes, minimize operational costs, ensure adherence to local regulations, and effectively manage limited trading options.

Technology and digitalization of currencies

Restricted markets, especially in Asia, have been been at the forefront of digital currency adoption. This proactive stance is driven by the potential of digital currencies to enable faster, more cost-effective, and secure cross-border transactions, leading these markets to explore various avenues like stablecoins, Central Bank Digital Currencies (CBDCs), and tokenized deposits.

Singapore is at the forefront, with a robust regulatory framework for single-currency stablecoins and the widespread use of XSGD, its Singapore dollar-pegged stablecoin. Hong Kong is also making strides, establishing a stablecoin licensing framework to become a global cryptocurrency hub and encouraging institutional adoption, with projects like HKDR (Hong Kong dollar-pegged stablecoin) in development. Indonesia has seen a surge in stablecoin adoption for cross-border transactions, with local stablecoins like IDRT and XIDR gaining traction and Bank Indonesia exploring a national stablecoin.

Central Bank Digital Currencies (CBDCs) are also being actively developed in Asia and other emerging markets to modernize financial systems and improve cross-border payments. China is a leader in CBDC development with its digital yuan (e-CNY). India's e-rupee is another CBDC pilot. Brazil has adopted the CBDC Drex although recent announcements suggest the coin is not going to utilise blockchain technology. Nigeria's eNaira was Africa's first central bank digital currency (CBDC).

Tokenized deposit pilots are flourishing in key Asian financial hubs like Hong Kong and Singapore. These initiatives coexist with stablecoins and CBDCs to enhance cross-border settlement, boost financial efficiency, and build more inclusive digital finance infrastructures while maintaining strong ties to traditional banks. Hong Kong's Project Ensemble, launched by the Hong Kong Monetary Authority (HKMA), and Singapore's Project Guardian, initiated by the Monetary Authority of Singapore (MAS), are prominent examples in this space.

The future of FX trading in restricted markets

Our work on developed markets has shown that 45% of the FX spot market actively seeks profit, while the remaining 55% is dedicated to business functions, hedging, or facilitating international investments rather than generating alpha. For those whose FX is predominantly operational, the same electronification paths now offer part of the solution by providing a framework for increased automation. This is more crucial than ever given the headwinds of rising regulation coupled with management fee compression.

FX trading in restricted markets is largely operational, with foreign investors often executing FX trades onshore as part of their capital market activities. For instance, a common regulation mandates that FX trades be linked to underlying local investments. Consequently, a significant portion of foreign exchange execution in restricted markets is handled through passive products, such as custodian standing instructions.

This does not necessarily mean that FX in restricted markets is not electronified. It simply means that the FX workflow is not fully automated and often requires manual steps. However, as regulations in restricted markets continue to evolve, the FX execution landscape adapts accordingly. For example, comprehensive FX outsourced solutions are gaining traction among restricted market currencies.

Hedging via onshore markets can result in lower costs, with onshore forward curves typically trading below the offshore curves. In other words, the onshore and offshore markets may not always be aligned and the ability to switch between them may be another area of cost efficiency. Conversely, for hedging via offshore markets, higher efficiency gains can be achieved with greater synchronization of prices between the onshore spot trade and the corresponding offshore hedging trade. Specialised FX providers could potentially help managers optimize their FX process on both fronts.

While buy-side and sell-side participants continue to collaborate to improve the FX execution space in these markets, local regulations are also evolving to become more supportive of onshoring of FX flows. For example, recent pushes by regulators, especially in South Korea, have improved accessibility to the onshore FX market. Asset managers in offshore locations can now access these markets through offshore banks during offshore hours. This should result in more competitive onshore FX markets and improved overall efficiency for domestic and foreign investors. **The question remains whether other restricted markets will follow the Korean model.**

Lastly, technology will continue to play a key role in restricted markets. For example, Asia, within these restricted markets, has been at the forefront of digital currency adoption. This proactive stance is driven by the potential of digital currencies to enable faster, more cost-effective, and secure cross-border transactions, leading Asia to explore various avenues like stablecoins, CBDCs, and tokenized deposits.

Appendix: A detailed overview of restricted markets

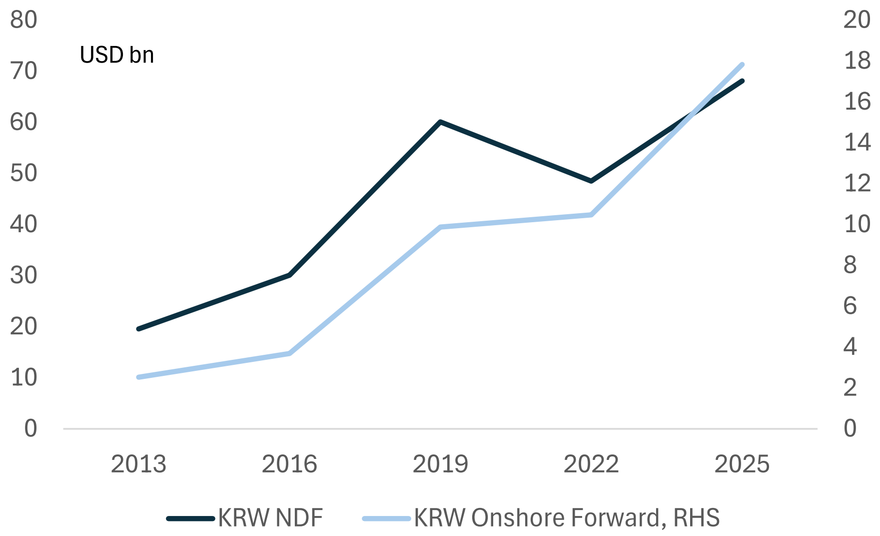
Korea: Non-residents access onshore markets via offshore banks

Korean regulators are introducing measures to open up the market for foreign investors. On the foreign exchange (FX) side, the Korean Ministry of Economy and Finance acknowledged that their FX market's restrictive structure had held back its growth, hindered stability, and made KRW-denominated assets less attractive. To address these issues, Korea announced significant reforms to align its FX market with global standards, with additional reforms (3 and 4) more recently. These include:

1. **Opened up the onshore interbank FX market to foreign financial institutions:** Non-residents can now trade Onshore KRW FX (Spot, Forwards and Swaps) via **Registered Financial Institutions (RFIs)**. *RFIs are Foreign FIs that are not based in Korea (Headquarters of foreign FIs, overseas branches of foreign FIs, and overseas branches of local FIs) that are allowed to directly participate in the onshore interbank FX market, if they obtain registration from the Korean authorities.*
2. **Extended onshore FX trading hours:** Local market trading hours have been extended till 2:00am local time (earlier close was 3:30pm). This allows asset managers to trade KRW without pre-funding transactions, reducing the costs and increasing the appeal of KRW assets.
3. **Facilitating RFI Transactions:** This will involve expanding the scope of RFIs to all transactions, including FX for current transactions like import/export payments. This expansion will enable RFIs to facilitate currency exchanges for multinational companies with headquarters or branches in Korea or abroad, and will also allow for salary remittance for their employees. Notably, individual fund-level identifiers are not required for trading, reporting, and monitoring for any FX, Bond, or Equity related transactions. Furthermore, Third-Party FX (TPFX) will be promoted by extending the settlement deadline for sending KRW funds to local custodian banks from 10:00 AM to 11:00 AM, and by broadening the range of institutions eligible for KRW Overdraft to include all institutions involved in securities transactions.
4. **Improving FX Brokerage Infrastructure**: Aggregators will be introduced to allow corporate end-users to select the most favorable prices through brokerage platforms.

According to "Foreign Exchange Trading Trends of Foreign Exchange Banks in 2025" the daily average foreign exchange trading volume (spot exchange and foreign exchange derivatives transactions) of foreign exchange banks last year totaled $80.71 billion. This represents a 17.0% increase from 2024 ($68.96 billion) and marks the highest level since the statistical reorganization in 2008. The Bank of Korea document further states that this is due to the continued impact of extended foreign exchange market trading hours in 2024, along with significant increases in transactions related to residents' overseas securities investment and foreigners' domestic securities investment.

Figure 8: Recent reforms have encouraged a shift from NDF into onshore forwards

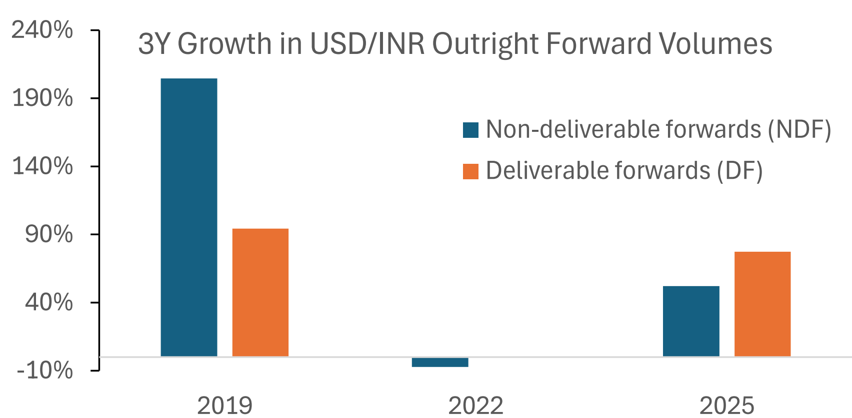


Source : Deutsche Bank, BIS

India

A 2019 Task Force on Offshore Rupee Markets recommended increasing onshore market participation and simplifying access without altering capital controls. In response, the Reserve Bank of India (RBI) made two key changes in 2020. First, **liberalized onshore risk management rules**. Moved to a principles-based approach, giving market makers more flexibility for hedging products (e.g., hedging anticipated exposures, equal treatment for residents/non-residents, expanded derivative access). **Second, allowed NDF market participation from GIFT City**: Indian and foreign bank units can now trade FX-settled NDFs with non-residents from GIFT City, an international financial services center operating outside India's main capital controls.This strategy has successfully redirected NDF flows onshore, enhancing RBI oversight and aligning deliverable and non-deliverable forward curves. The latest RBI draft requires reporting of all OTC foreign exchange derivative contracts involving INR, undertaken globally by the related parties of the Authorised Dealer Category-I (AD Cat-I) banks.

Figure 9: RBI reforms have potentially led to more flows onshore



Source : Deutsche Bank, BIS

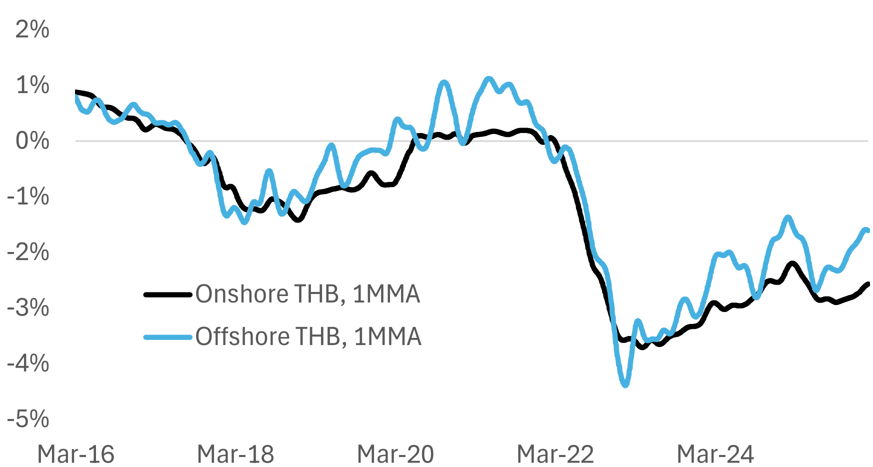
Thailand

Thailand's Baht (THB) has a unique currency control system, leading to a split between onshore and offshore THB forward curves due to limitations on non-resident THB balances and documentation for onshore trading.

In 2021, Thailand launched the Non-Resident Qualified Company (NRQC) scheme to integrate these markets. This scheme allows non-resident corporations with Thai trade and investments to access onshore THB liquidity and hedge freely. Further relaxations were announced, effective December 1, 2025, allowing non-residents to conduct transactions without underlying documents and maintain unlimited THB deposits, making the NRQC scheme generally applicable without case-by-case approval.

The NRQC scheme provides economic benefits, as the onshore THB curve has historically traded **35 basis points** below the offshore curve, and the onshore market offers greater depth in longer-dated forwards. This incentivizes corporates to shift hedging onshore, resulting in cost savings and increased efficiency, as also highlighted by our colleagues in Asia Thematic Strategy Notes: Onshoring FX flows: regulation is turning the market tables.

Figure 10: The onshore THB curve generally cheaper and stable



Source : Deutsche Bank, Bloomberg Finance LP

China

We highlighted earlier China's internationalization follows an idiosyncratic route. Chinese authorities engineered a system allowing a substantial pool of Renminbi to accumulate and be freely traded and delivered offshore, all while maintaining capital controls. This strategy has resulted in a tripartite structure for the Renminbi forward market: an onshore market (dating to 2006), an offshore NDF market (established in the 1990s, CNY NDF), and an offshore deliverable, or CNH, market (emerging in 2010, CNH DF).

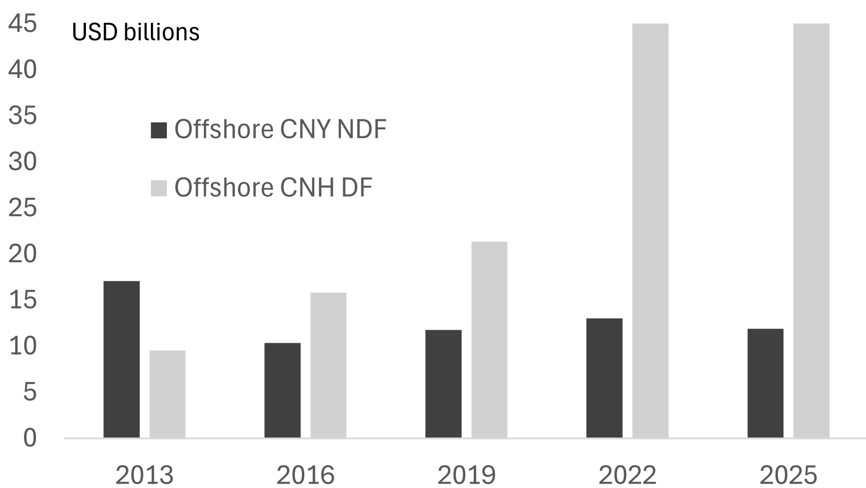
China is actively internationalizing the RMB by promoting its use in cross-border trade and investments. The PBOC has focused on improving cross-border RMB settlement through:

- **International cooperation:** Expanding bilateral local currency-RMB swap lines with over 40 countries and establishing an RMB liquidity agreement for central banks.

- **Enhanced infrastructure:** Continuously building out the CIPS payment platform (with 193 Direct and 1573 Indirect Participants across 124 countries as of December 2025, up from 19 Direct Participants and 176 Indirect Participants from 50 countries in 2015). Extending onshore FX market trading hours to 3 AM Beijing Time (effective January 3, 2023) to cover global time zones.
- **Corporate empowerment**: Initiatives like cash pooling programs for MNCs and increased flexibility in using Panda bond financing proceeds.

According to our latest EM handbook, China, with its rapidly growing bond market (now the world's second-largest), is progressively opening up to foreign investors. This involves initiatives like promoting Dim Sum bonds, encouraging local-currency settlement, and testing the digital RMB, ultimately granting corporations increased flexibility in their funding and hedging strategies.

Figure 11: CNH has largely surpassed CNY NDFs



Source : Deutsche Bank, BIS

Other Asian markets

Malaysia shifted currency flows from offshore to onshore in 2016 by prohibiting licensed banks from facilitating NDF trades, following USD/MYR volatility. This policy reduced NDF liquidity and speculative activity, forcing hedging onshore. The BNM supported this with a Dynamic Hedging Framework, liberalizing onshore hedging for non-residents and significantly boosting onshore forward liquidity. Further foreign exchange policy liberalizations in April 2021 granted exporters greater flexibility in managing foreign currency proceeds, eased FX hedging restrictions for residents, and enhanced capital market efficiency by simplifying onshore foreign exchange activities.

Indonesia introduced the IDR-settled Domestic Non-Deliverable Forward (DNDF) in 2018, traded onshore and auctioned daily by Bank Indonesia. While not primarily designed to shift offshore hedging onshore due to documentation requirements,the DNDF serves a critical function: it maintains hedging market access, provides an additional tool for BI intervention, and allows the onshore DNDF curve to influence offshore NDFs, rather than being solely driven by offshore volatility during stress. The DNDF has effectively contained offshore NDF points during global turmoil, enabling non-residents to continue hedging economically and preventing forced asset sales.

Taiwan allowed partial integration between onshore and NDF markets, as Taiwanese banks are permitted to participate in the NDF market up to 20% of their net open FX positions. They are restricted to trading NDFs solely with locally registered banks and their offshore sister branches. Onshore banks are prohibited from booking NDFs with corporations. Taiwan CBC also imposes limits on nonresident investment in local currency bonds and mandates onshore currency transaction reporting requirements.

Brazil

Brazil operates extensive and developed onshore derivatives exchanges for an emerging economy, handling FX, interest rate, stock, and commodity instruments. Brazilian FX futures and options are non-deliverable, settling in domestic currency.

Latest BIS study shows distinctive nature of the BRL FX market, noting a comparatively low proportion of its FX derivatives trading offshore. The BRL is unique as a significant volume of its FX derivatives trading occurs via futures on the Sao Paulo exchange, a characteristic less prominent for other currencies. Crucially, the limited restrictions on non-resident participation in BRL futures contribute to this onshore liquidity concentration.

This market structure, characterized by predominant exchange-traded, non-deliverable transactions, evolved due to a combination of factors including tendency to economic instability that generated demand for hedging. Historical high and volatile inflation, followed by increased foreign currency borrowing post-Real Plan, consistently drove the need for risk management tools in both interest rate and FX markets.

The Brazilian legal and regulatory framework significantly influences this derivatives market. Constraints on over-the-counter (OTC) trading and a tax system favoring net profit/loss calculations on exchanges encourage trading migration to these platforms. Additionally, strict limitations on foreign currency use and trading, such as restricted FX spot market access and the Brazilian Real's non-convertibility outside the country, necessitate non-deliverable, local currency-settled derivatives as substitutes for cash transactions.

Last year, Brazil announced substantial tax changes affecting financial and capital markets taxation – the IOF tax – which increased the levy on financial transactions.

Colombia

The Colombian FX market has a liquid, competitive, and efficient spot FX market by international standards, contrasted with a hedging market predominantly characterized by Non-Deliverable Forwards (NDFs). As per the IMF report, the spot FX market has relatively high trading volumes, with daily average inter-dealer turnover around USD 750 million and dealer-to-customer volume around USD 600 million, largely driven by nonfinancial customers, domestic pension funds, and foreign institutional investors. The interbank FX market is competitive, with tight bid-ask spreads.

Colombia's FX turnover has increased, but a significant portion of this surge is due to offshore NDF trading, keeping its overall turnover lower than the average across emerging markets. Between 2016 and 2022, offshore trading averaged around USD 7 billion, significantly surpassing combined onshore and cross-border trading of approximately USD 4 billion. This rapid offshore growth contrasts with the slower increase in onshore and cross-border activity. The prevalence of NDFs as the primary hedging instrument stems from historical capital controls and regulatory constraints inhibiting non-residents' COP settlement, which in turn hinders the development of deliverable forwards and swaps. Furthermore, a financial transaction tax renders deliverable derivatives more expensive than non-deliverable ones.

Argentina

According to the IMF report, non-resident investors are no longer subject to a minimum holding requirement in order to participate in local debt markets, as well as to the cross-restriction between the official and parallel markets. Additionally, the requirement of a 48-hour advance notice for large FX purchases has been removed. The BCRA has also issued a new 3-year FX-denominated bond (BOPREALs) targeted at firms with legacy commercial debt and dividends accumulated during the capital control years, providing greater predictability about the resolution of these legacy external obligations. These measures are expected to catalyze further foreign investments.

Frontier markets

The term "frontier markets," coined by the International Finance Corporation in 1992, refers to markets that are relatively smaller with foreign investor access limits but offer promising investment opportunities. The MSCI Frontier Markets Index includes large and mid-cap companies across 28 such countries, with top constituents like Vietnam, Romania, Morocco, Kazakhstan, and Slovenia. Other notable frontier countries include Nigeria, Ghana, Kenya, Uganda, Zambia, Uzbekistan,Uruguay, Cost Rica and Pakistan.

Frontier Markets are characterized by their early stages of capital market maturity, often lacking financial depth and a shallow domestic investor base. Many of these markets fall below the emerging market median in terms of overall financial development and the depth of their local financial markets. This lack of financial depth also translates into challenging local market liquidity conditions, where bid-offer spreads and the price impact of trades are significantly larger compared to other emerging markets.

Direct local spot settlement in these markets can be illiquid, leading investors to use Non-Deliverable Forwards (NDFs) for exposure without local accounts. Accessing these markets, particularly for African currencies, often requires specialised institutions and local correspondent networks.

For a more detailed discussion of the regulatory framework please see our latest EM handbook.

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Appendix 1

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